INTERNSHIP PROJECT REPORT · PRESENTATION

SaaS Subscription Churn Prediction

Machine Learning and Interactive Analytics

Pratham Rangoonwala

Naviotech Solution Pvt. Ltd.

25/10/2026

CONTEXT

Why churn, and what this project does

Objectives

- Clean and combine five SaaS tables into **one row per account**
- Describe how churn varies by plan, country, industry, tenure and behaviour
- Train and compare classifiers; choose model and threshold **without touching the test set**
- Present findings, results and a risk lookup in a **six-tab Streamlit dashboard**

Scope boundaries

- **Synthetic dataset** (Kaggle) - results describe this data only
- **Associations, not causes** - no causal claims
- **No operational claims** - no deployment, real-time scoring or alerts
- **Validation ≠ test** - metrics are always labelled by split

DATASET

RavenStack: five related tables, 500 accounts

accounts

500 × 10

profile + churn_flag

key: account_id

subscriptions

5,000 × 14

MRR, plan, flags

key: account_id

feature_usage

25,000 × 8

usage, errors, beta

key: subscription_id

support_tickets

2,000 × 9

response, satisfaction

key: account_id

churn_events

600 × 9

date, reason, refund

key: account_id

Target: churn_flag

110 churned · 390 retained

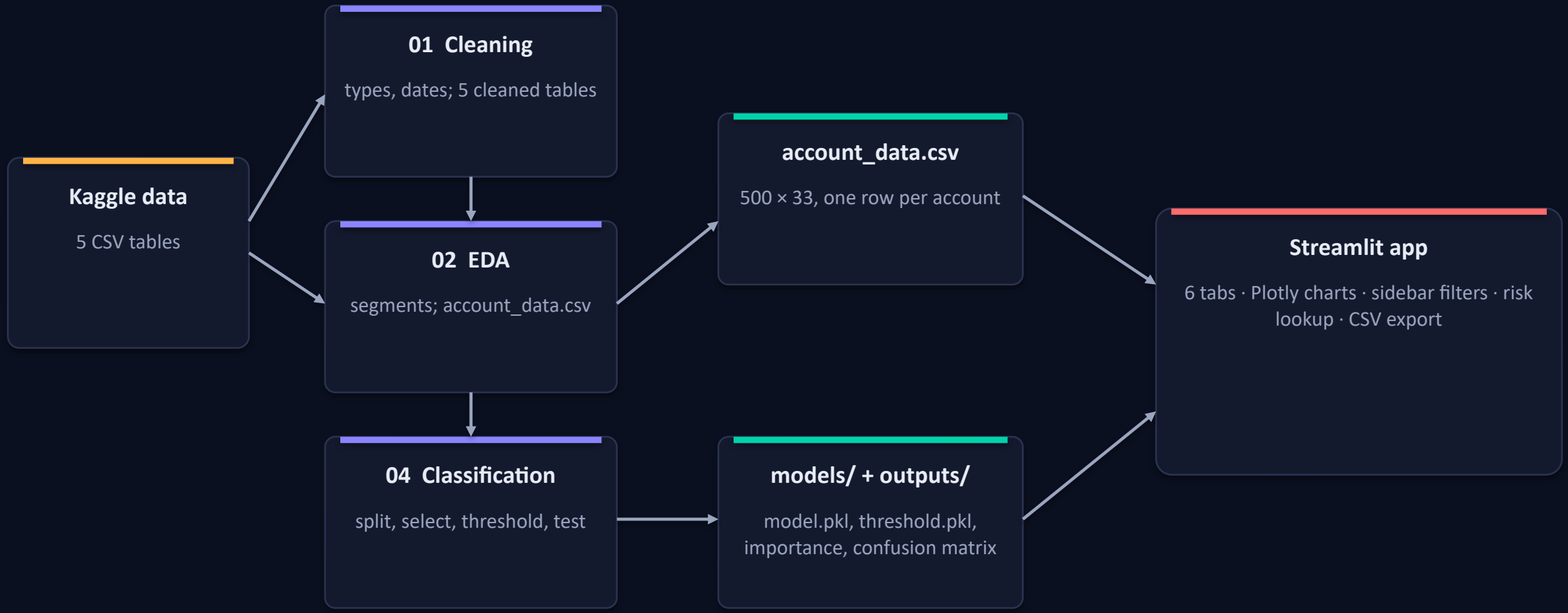
22.0% churn rate. The flag is used as supplied; how it was derived is not visible in the project code.

Source and data quality

- Kaggle: SaaS Subscription & Churn Analytics (River @ Rivalytics)
- Public descriptions call it **synthetic**
- No duplicate rows; missing values only in end_date, satisfaction_score and feedback_text

SYSTEM DESIGN

From raw CSVs to an interactive dashboard



Notebook 03 is an earlier iteration (80/20 split) superseded by notebook 04 - not used for any result in this deck.

DATA PREPARATION

Features use only what was known before the prediction date

Prediction date per account

Retained → 2024-12-31 (end of the data)

Churned → first churn event − 30 days

Subscription, usage and ticket records are filtered to dates ≤ the prediction date before aggregation. Tenure is measured at that date.

Account profile

plan tier, industry, country, seats, trial, tenure

Subscription

avg MRR, upgrades, downgrades, auto-renew

Product usage

usage, duration, errors, beta usage

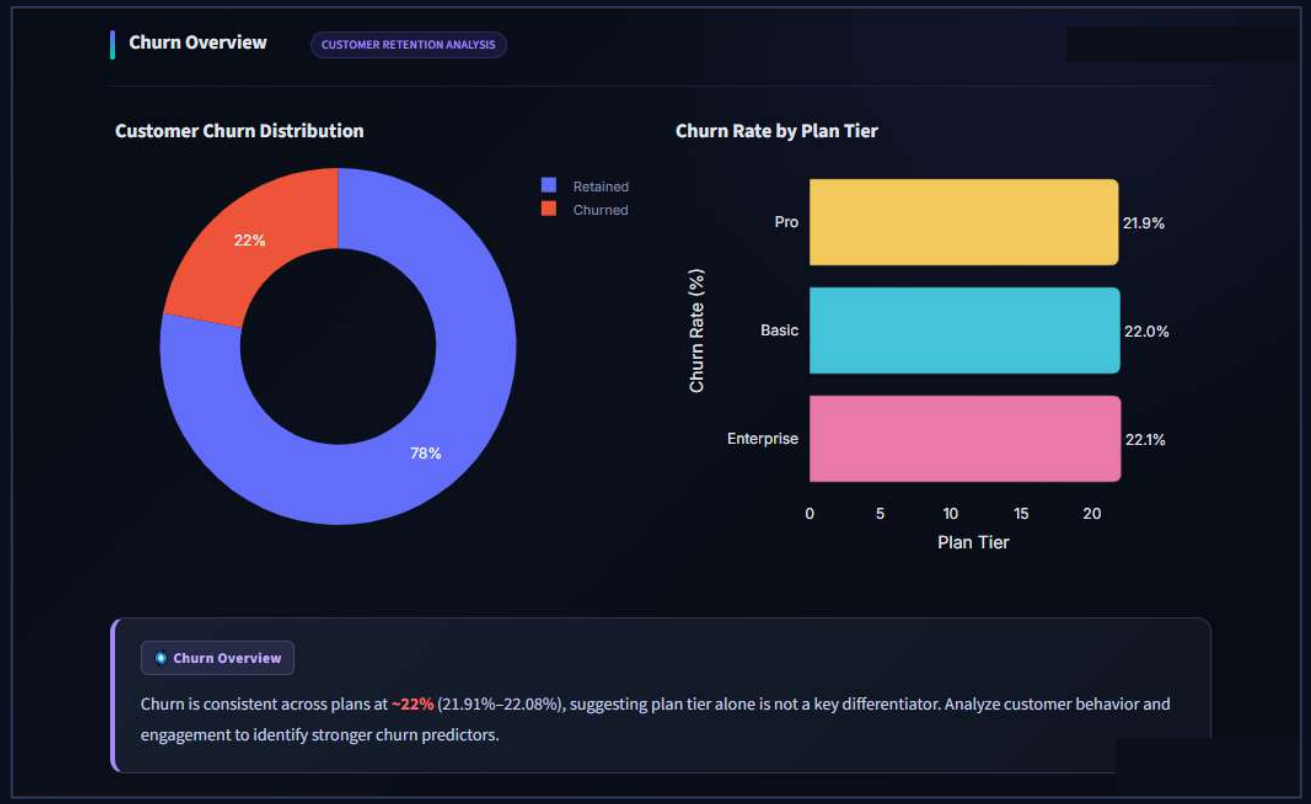
Support

tickets, resolution, response, satisfaction, escalations

19 features (31 after encoding). Excluded as leakage risks: churn_date, reason_code, refund_amount_usd, feedback_text.

EXPLORATION

Churn is flat by plan; some segments stand out



~22%

across all plan tiers (21.9–22.1%)

31.0%

DevTools churn (35 of 113)

32%

Germany (8 of 25); US has most churned: 68

25.8% vs 21.1%

trial vs non-trial accounts

Event-referred accounts: 30.2% (29 of 96).
Tenure groups: 18.7% / 23.5% / 31.2% - but tenure is measured to the churn date for churned accounts, so this reflects how tenure is defined.

Screenshot: Churn Overview tab of the dashboard.

STREAMLIT DASHBOARD

One page, six tabs, sidebar filters



1 Churn Overview

2 Geography

3 Tenure & Engagement

4 Subscription & Behavior

5 Classification Model

6 Prediction & Risk

- Filters: **Country, Plan Tier, Industry**; four KPI cards: 500 customers, 110 churned, 22.0%, $2,264.18 average MRR
- Charts built with **Plotly Express**; custom dark theme (CSS)
- Insight boxes and recommendation cards are **static text** written in the app code

DASHBOARD VIEWS

Segment and behaviour charts



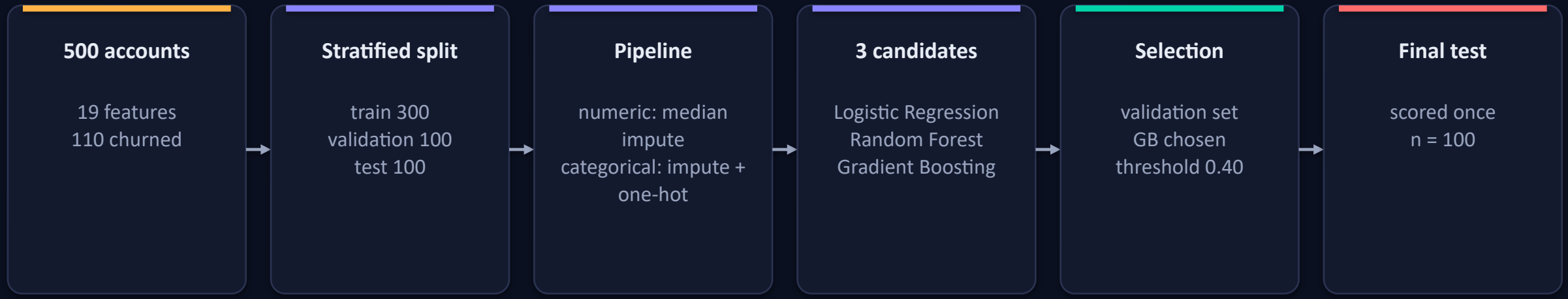
Geography: churn rate by country and plan. Germany highest (32%); several cells rest on very few accounts.



Subscription behaviour: upgrades 1.06 vs 1.05 and downgrades 0.44 vs 0.43 - nearly identical.
Average MRR: retained $2,315.30 vs churned $2,082.92.

MODELLING

Select on validation, evaluate once on test



- Gradient Boosting: 150 trees, learning rate 0.05, depth 2, min leaf 5 (fixed settings; no tuning search recorded)
- Threshold scanned 0.20–0.70 in steps of 0.05; best validation F1 = 0.757 at 0.40 (0.45 ties)
- The model is fitted on the **training split only**

MODEL SELECTION

Validation results (n = 100, cutoff 0.50)

VALIDATION SET

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.710	0.370	0.455	0.408	0.686
Random Forest	0.850	0.818	0.409	0.545	0.810
Gradient Boosting	**0.900**	**0.929**	**0.591**	**0.722**	**0.805**

Why Gradient Boosting

Highest accuracy, precision, recall and F1 on validation; ROC-AUC close to Random Forest.

Cross-check (not used to select)

5-fold stratified CV on all 500 rows: accuracy 0.926, F1 0.810, ROC-AUC 0.916 at a 0.50 cutoff.

FINAL EVALUATION

Held-out TEST results (n = 100, threshold 0.40)

TEST SET

Accuracy	Precision	Recall	F1 score	ROC-AUC
0.900	0.800	0.727	0.762	0.876

	Predicted retained	Predicted churned
Actual retained	74	4
Actual churned	6	16

Not comparable with the validation table: different accounts and a different threshold.

How to read it

- 16 of 22 churners found; 6 missed
- 16 of 20 flagged accounts were real churners
- Only 22 churners: bootstrap 95% interval for recall ≈ 0.52–0.90 (supplementary)

MODEL DRIVERS

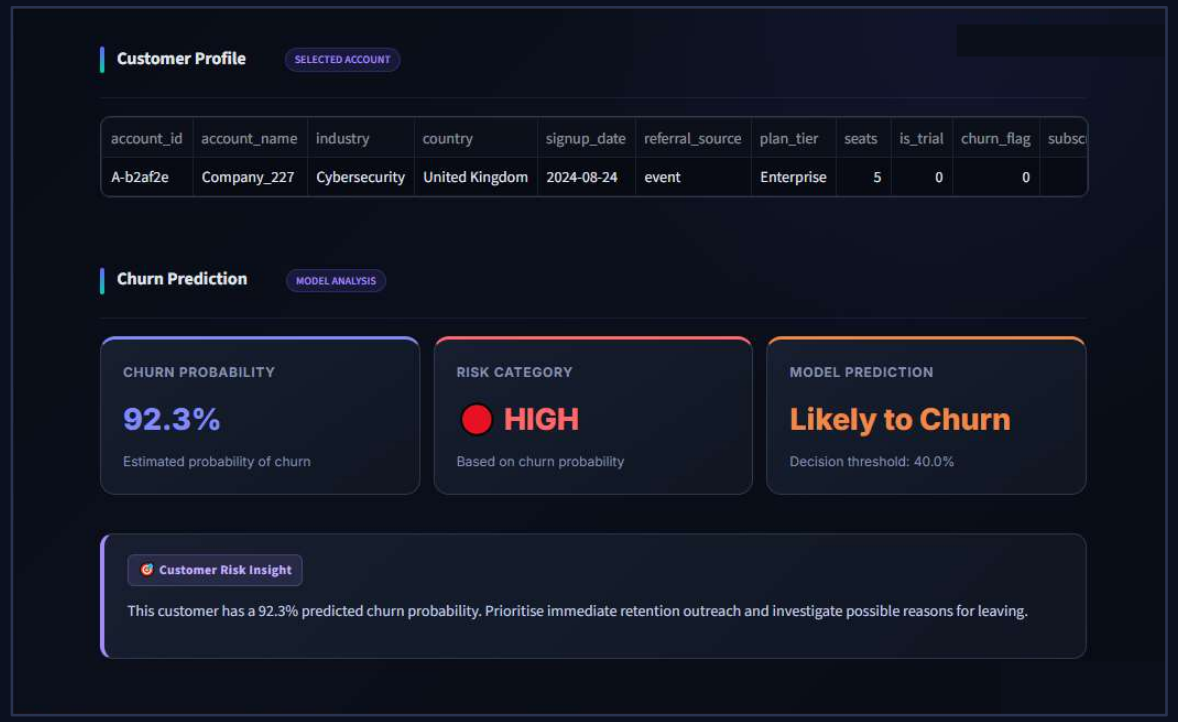
What the Gradient Boosting model relies on

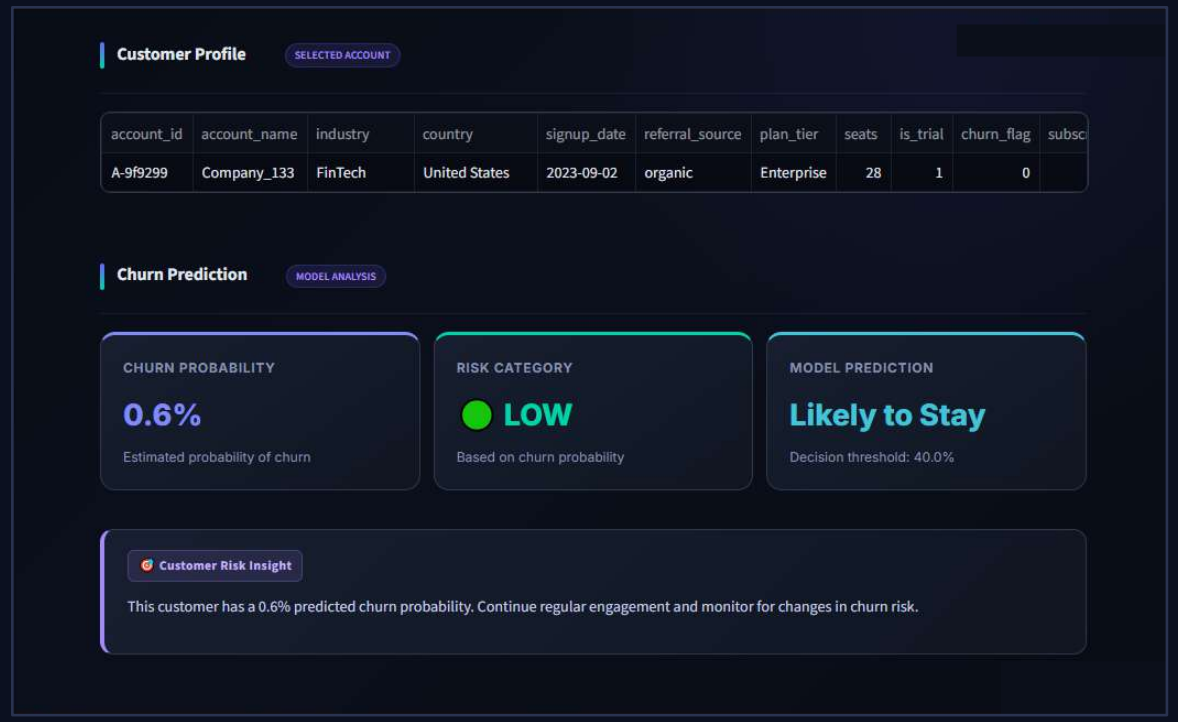


- **Avg MRR 52.8%**, Auto Renew 15.2%, Tenure 14.6% - together 82.6%
- Importance shows what the trees split on, **not what causes churn**
- Caution: avg MRR is missing for 62 of 110 churned accounts and 0 of 390 retained - part of its importance may reflect that pattern (next slides)

PREDICTION & RISK TAB

Per-account churn probability and risk band





HIGH

probability ≥ 0.60

MEDIUM

0.40 ≤ probability < 0.60

LOW

probability < 0.40

Illustrative lookups only: the tab scores whole-history features from account_data.csv, which differ from the training windows, and ~60% of accounts were training rows.

LIMITATIONS

What the numbers do and do not show

Small test set

22 churners in 100 accounts. One account moves recall ~4.5 points; intervals are wide.

Missing features are informative

11 of 22 test churners have no avg MRR (no churn event → no prediction date). A rule 'avg MRR missing → churn' alone scores precision 1.0, recall 0.50. On the 89 test accounts with real values: AUC ≈ 0.76, recall ≈ 0.45.

Dashboard scoring mismatch

Prediction tab uses whole-history features; the model was trained on pre-prediction-date windows.

Data & app caveats

Synthetic data; churn_flag derivation not shown; model KPI cards, validation chart and insight text are hard-coded.

Reported test metrics are the project's own results; the supplementary figures come from a reproduction and diagnostics run while documenting it.

SUMMARY

Conclusion and next steps

What was delivered

- End-to-end pipeline: five tables → account-level data → model → dashboard
- Plan tier is not associated with churn; DevTools, event referrals, trial accounts and Germany show higher rates
- Gradient Boosting, threshold 0.40: **test accuracy 0.900, recall 0.727, ROC-AUC 0.876** (n = 100)

Suggested next steps (not yet done)

- Assign prediction dates without relying on the label; re-evaluate with and without missing indicators
- Score the dashboard with training-consistent features
- Repeated CV and confidence intervals
- Generate app figures from saved result files
- Validate on a real, non-synthetic dataset